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9. (a) A student tested compounds **A**, **B** and **C** to identify them.

Her observations are recorded in the tables below.

Compound **A**	Observation	Ion present
Flame test	apple-green flame	barium
Add silver nitrate solution	white precipitate	chloride

Compound **B**	Observation	Ion present
Flame test	lilac flame	
Add silver nitrate solution	cream precipitate	

Compound **C**	Observation	Ion present
Flame test	brick-red flame	
Add silver nitrate solution	yellow precipitate	

(i) Use your knowledge of the tests for ions to complete the tables for compounds **B** and **C**. [3]

(ii) Write the chemical formula for compound **A**. [1]

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(b) When silver nitrate solution reacts with sodium chloride solution to produce a white precipitate the following reaction occurs.

$$AgNO_3(aq) + NaCl(aq) \longrightarrow NaNO_3(aq) + AgCl(s)$$

Put a tick (✓) in the box which shows the ionic equation for the formation of the precipitate. [1]

$Na^+(aq) + NO_3^-(aq) \longrightarrow AgCl(s)$	☐
$Ag^+(aq) + NO_3^-(aq) + Na^+(aq) + Cl^-(aq) \longrightarrow NaNO_3(s) + AgCl(s)$	☐
$Ag^+(aq) + NO_3^-(aq) \longrightarrow AgNO_3(s)$	☐
$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$	☐
$Ag^+(aq) + Cl^-(aq) \longrightarrow NaNO_3(s)$	☐

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(c) When silver carbonate is heated it breaks down to give silver metal and carbon dioxide and oxygen gas.

$$2Ag_2CO_3 \longrightarrow 4Ag + 2CO_2 + O_2$$

Calculate the mass of silver that would be produced from heating 13.8 g of silver carbonate. [3]

$M_r(Ag_2CO_3) = 276$ $A_r(Ag) = 108$

Mass = ... g

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END OF PAPER